Group 8: Brealin Redecker, Xingjian Murong, Yiyi Su

NATIONAL BOOKSELLERS INC FINAL REPORT

Business Background:

NBI buys books and compact discs from large distributors and resells to bookstores, computer stores, and other retailers. The company is led by Bill Tyler the CEO, Maire Corbit the VP, Richard Gonzalez the Director of Marketing, Carol Bailey the Accounting Manager, Michael Nakatani the Warehouse Operations Manager, and John Lackritz the Operation/IS Manager. Currently, the system is facing some issues including: the vendor of the current proprietary system offers limited support, upgrades to the operating system are not frequent enough, lacks adequate ad-hoc reporting features, slow and unstable, performance is inadequate for current number of users, and users expected to increase. And the company wanted to conduct the Information System Upgrade Project to replace the company's old system, and meet the requirements including: Software apps should run on any hardware, be online within 4 months, must be affordable, NBI gets rights and can modify the code, and data must be exportable to existing packages until conversion is successful.

Scope:

The NBI Information System will process orders, process purchasing and updating inventory received from vendors, and process returns.

Functional Requirements:

- Process customer orders.
- Handle products purchasing from vendors/publishers.
- Process customer orders return requests with refund.
- Store customer and publisher/vendor information.

Non-Functional Requirements:

- Support the communication for at least 100 users across the site in California and the site in New York.
- Support documents printing.
- Support online and business phones for customer services.
- Restrict system access based on user roles.
- Provide customer purchasing statements every 30 days.

System Analysis:

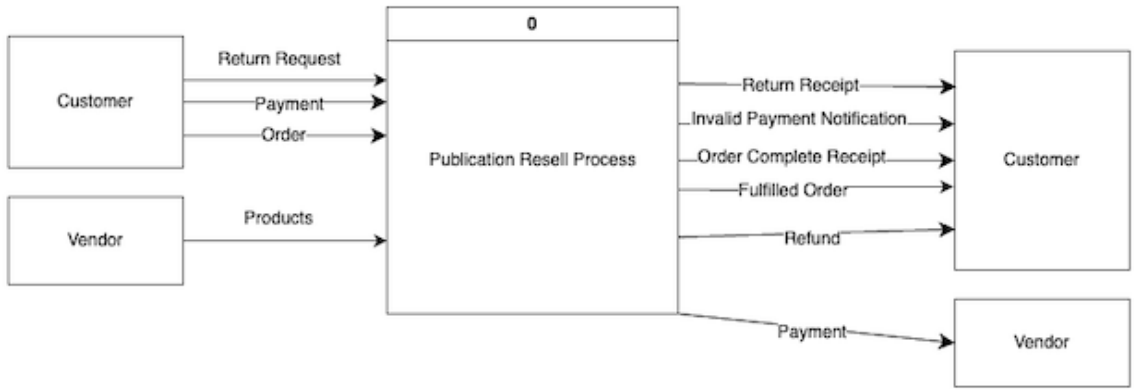
Business assumptions:

- A receipt will be sent to the customer for each order/return request; a failure notification will be sent to the customer if the payment is invalid.
- Each transaction will update the inventory.

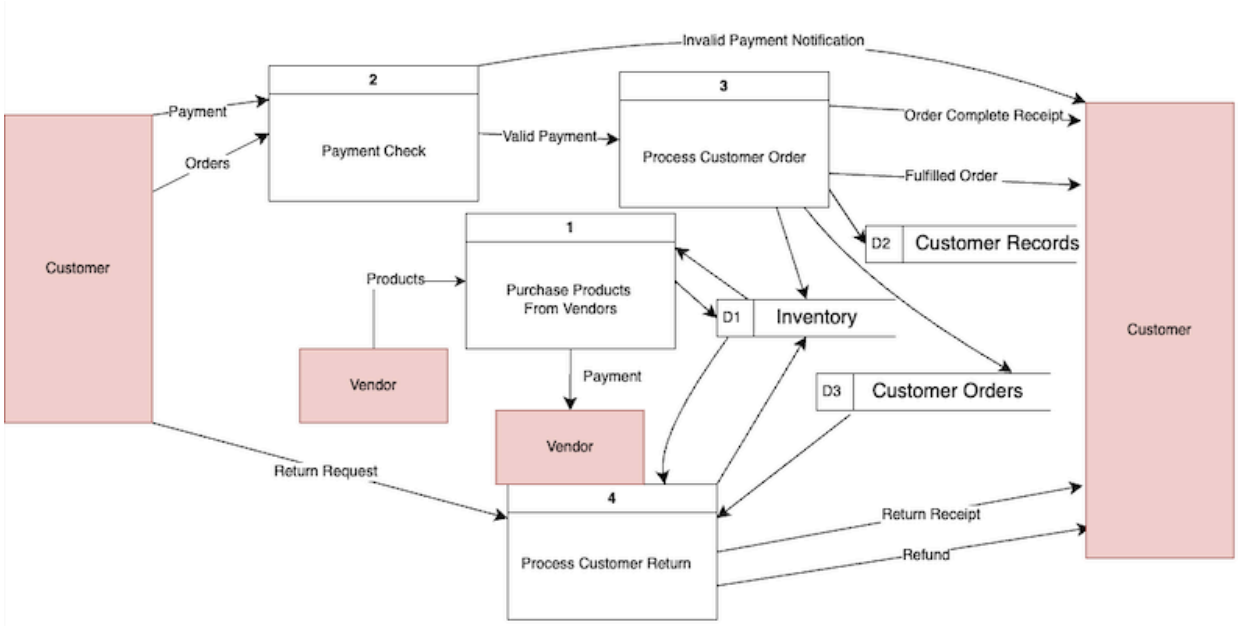
- A new customer record will be made when processing each order, and the order item details will be stored into the customer order datastore. If there is a repeating customer, new details will be updated in the customer order datastore.
- The information in the customer order datastore will be referenced when processing return requests.
- A refund will be sent to the customer after processing the return request.

DFD:

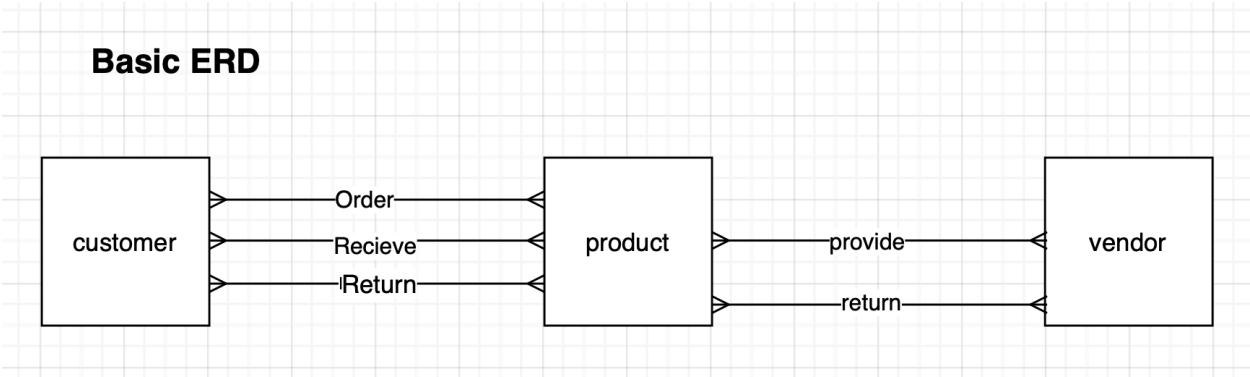
Context DFD

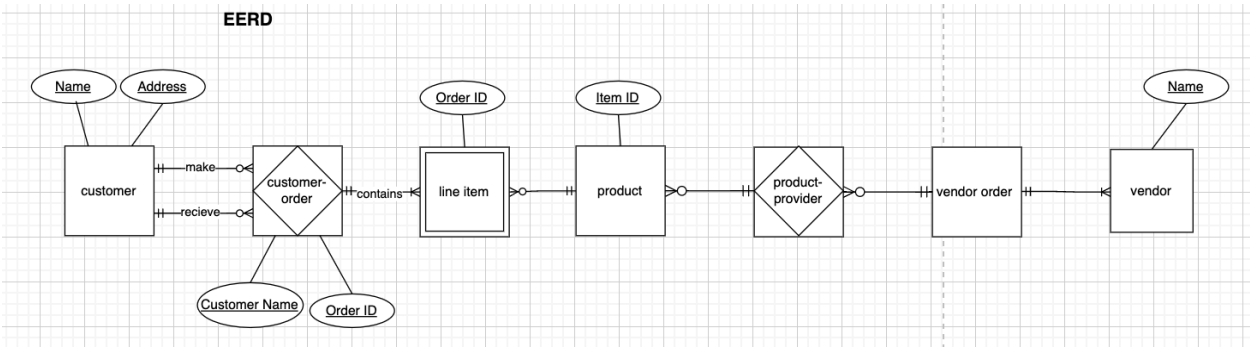


Level-0



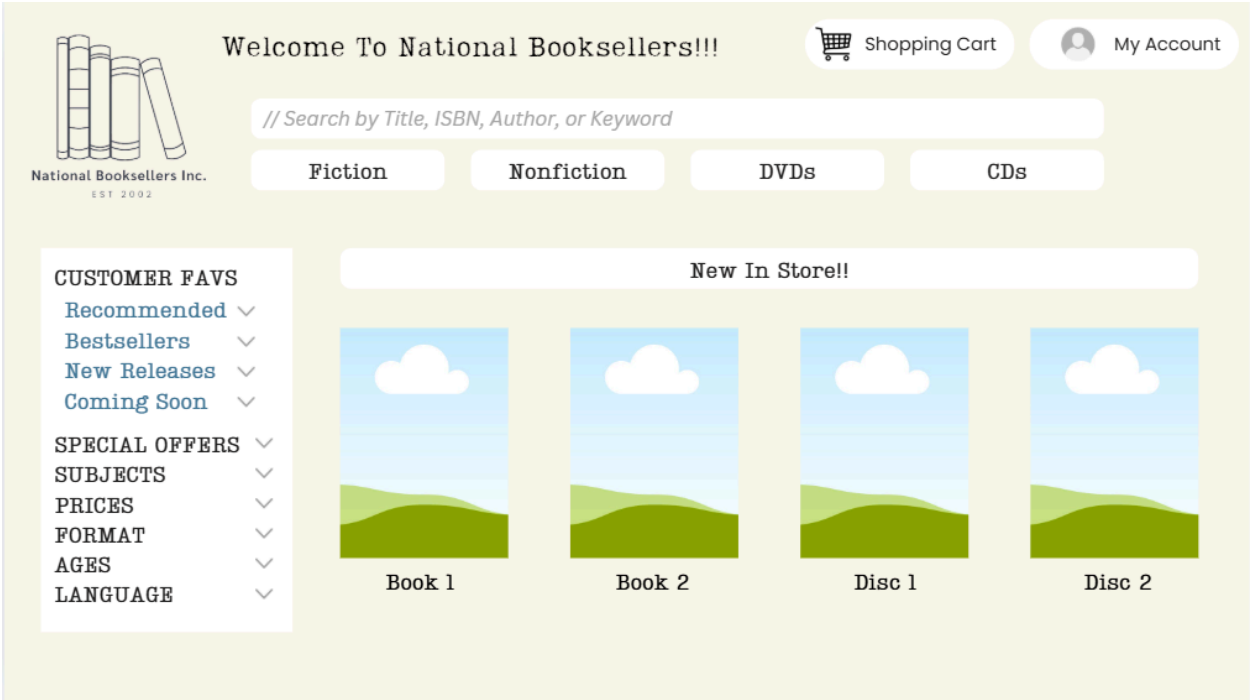
ERD:



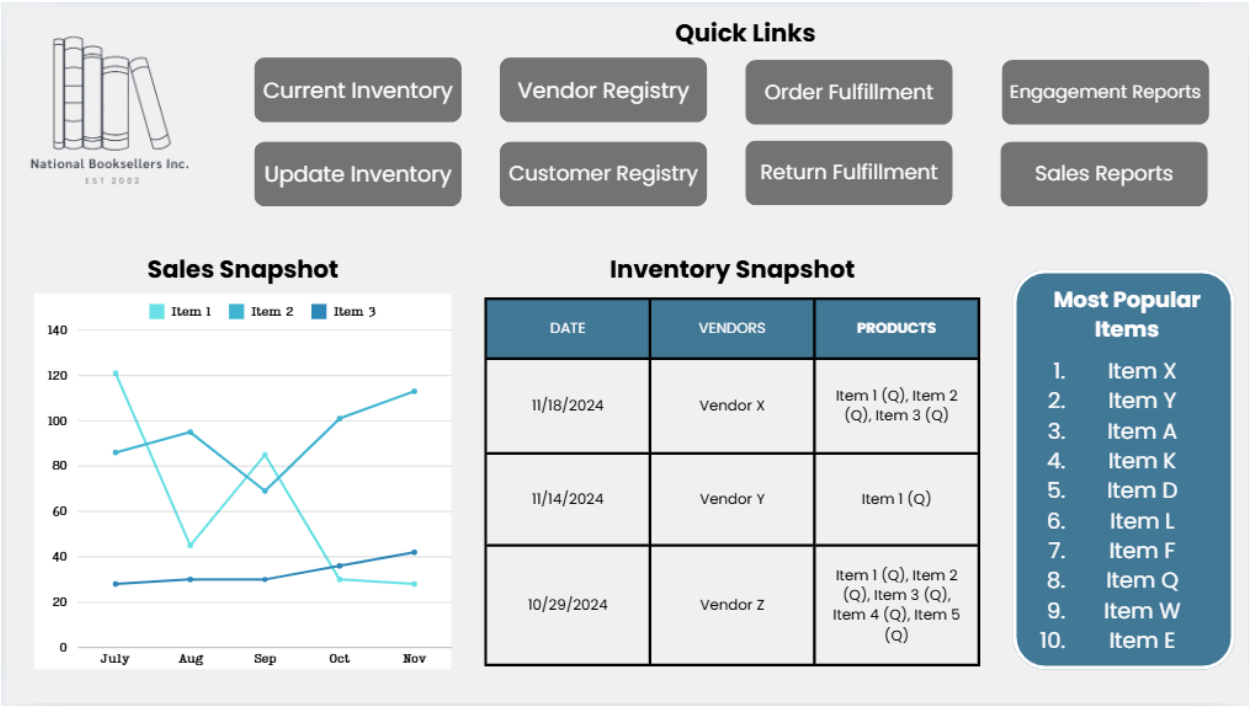


User Interface Design:

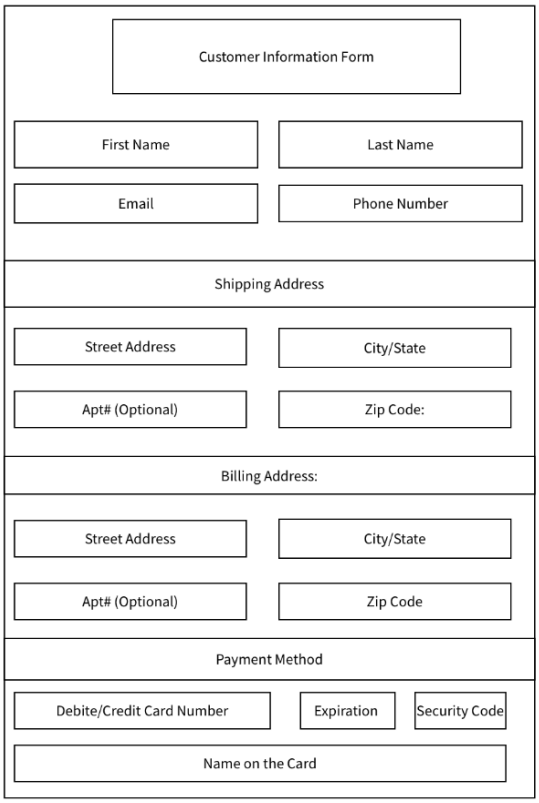
- The landing screen for customers

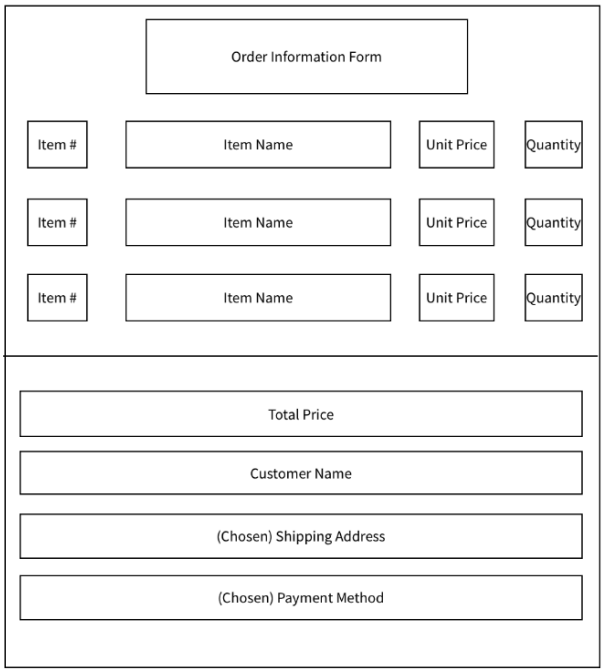


- The landing screen for NBI manager



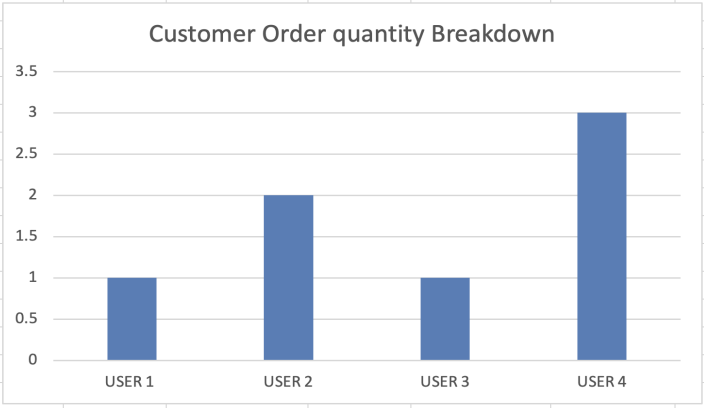
- Input Forms



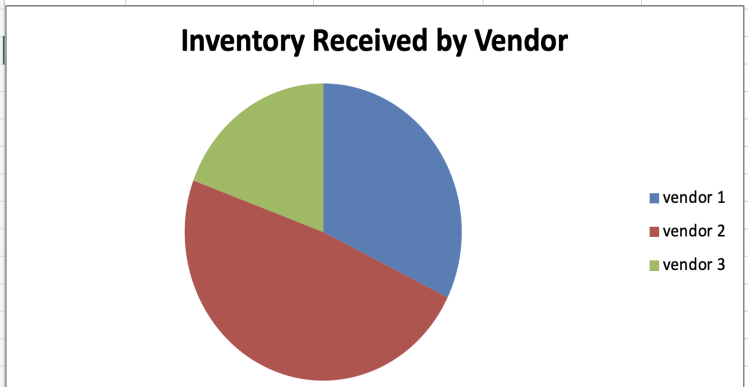


- **Output Reports**

Customer N	Order ID	Product Name	Quantity	Order Date	Order Status	Refund Amount
USER 1	ORD001	Book A	1	2024-11-01	Delivered	-
USER 2	ORD002	Book B	2	2024-11-02	Pending	-
USER 3	ORD003	Book C	1	2024-11-03	Returned	$15
USER 4	ORD004	Book D	3	2024-11-04	Delivered	-



Vendor Name	Vendor Order ID	Product Name	Quantity Received	Transaction Date	Status
vendor 1	VEND001	Book A	100	2024-10-20	Completed
vendor 2	VEND002	Book B	50	2024-10-21	Pending
vendor 3	VEND003	Book C	75	2024-10-22	Completed
vendor 4	VEND004	Book D	30	2024-10-23	Returned



Reflections & Insights (This needs to be about 2 pages):

- What did you learn as an analyst in this project?
- Include problems, difficulties, lessons learned, and experience gained.
- Suggestions to others or in future projects.

Brealin Redecker:

Throughout this project, I gained a deeper understanding of the responsibilities and challenges associated with being a systems analyst. The experience underscored how crucial attention to detail, thorough communication, and problem-solving skills are in the development and implementation of effective information systems.

One of the most difficult aspects was distinguishing between functional and non-functional requirements. Initially, it was unclear which details belonged in each category—functional requirements describe specific tasks the system must perform, like processing orders or handling returns, while non-functional requirements focus on how the system performs these tasks, such as system performance or user support. Over time, I realized that it takes practice and repetition to recognize these distinctions, and it's important not to be too hard on ourselves when it's unclear at first. Another challenge was understanding the appropriate level of detail for Data Flow Diagrams (DFDs) and Entity-Relationship Diagrams (ERDs). I struggled to determine what information was too broad or too specific, especially as we moved between different levels of the diagrams. Through iterative feedback and collaboration, I learned to focus on the Context DFD in early stages and gradually add more detail as we built the Level 0 DFD.

Being a system analyst and going through the lifecycle with a project is not as straightforward as it seems. Building DFDs and ERDs is not just like speaking another language, it is also an art form that requires time, effort, and patience to understand. This experience showed me how important it is to take system design step by step and make improvements based on feedback. I realized that understanding and getting things right takes practice, and it's really helpful to ask for feedback early and often. For future projects, I would suggest starting with simple ideas, adding more details as you go, keeping track of any assumptions, and working closely with the team to stay on the same page.

Yiyi Su:

This project allows me to have hands-on experience to handle a more comprehensive case than those class activities, from understanding the case to the final design. It is a great experience to shape my skill as a system analyst, and practice in team work. We analyzed the business case to specify the scope of the project, utilized the knowledge we gained from the class

to develop DFD and ERD to visualize the system process, then designed UI. This entire process is such a great experience to deepen my understanding of the planning phase in SDLC. And also a chance for us to feel the struggle in a position of system analyst.

At the beginning we had some difficulties to find out the scope and functional requirement. We had to redo the first part of this project, and thanks to the help from Professor Zhang, we finally understood the requirements of the system with a clear scope. As we overcame this challenge, the rest of the project went way more smoothly. Even though the process of developing ERD and DFD was still time consuming, we can always reference the scope and functional requirements to choose the most appropriate approach.

By finishing this project, I learned that the most important part of a project is to have a clear scope and requirement, as a guide for us to start. And I will keep that in mind for my future projects.

Xingjian Murong:

This project has been a valuable learning experience for me as a systems analyst. It gave me the chance to apply what I've learned in class to a real-world situation. One of the most important things I learned was how to clearly define what the system needs to do (functional requirements) and how it should perform (non-functional requirements). This wasn't easy at first, but through teamwork and feedback, I got better at it.

Working on diagrams like the Data Flow Diagram (DFD) and Entity-Relationship Diagram (ERD) was also a key part of the project. These helped me understand how information moves through the system and how different parts connect. It took a few tries to get the right level of detail, but the process taught me the importance of step-by-step improvement.

Teamwork was another big part of this experience. Communicating clearly with my teammates and dividing tasks helped us stay organized and finish on time. We also learned the value of asking for feedback and using it to improve our work.

Overall, this project taught me that designing a system requires patience, attention to detail, and good collaboration. These lessons will definitely help me in future projects.

Interaction Worksheet

Date	Attendees	Ways of Communication	Appx. Duration/ Length	Topic	Major Decision	Other Comments
(e.g.:	Prof, teammate(s). Note who is missing and why.	In person, email, IM, Chat, online meeting, phone, other	Minutes, hours, pages	Organizational meeting, redefine system boundary	Meeting days/times and file sharing methods, leader and note taker; new statement of system objectives	Tasks to do in next meeting; task allocation to individual members
10/9/2024	Brealin, Yiyi, Xingjian	In Person	12 minutes	Establish Team and Communication Plan. Decided on our case.	We shared this document amongst the group and created a folder for shared collaboration. We created a group chat. We decided on National Booksellers Inc.	Need to coordinate time to meet. Begin project proposal.
10/14/2024	Brealin, Yiyi, Xingjian	Group Chat (IM)	10 minutes	Begin work on Project Proposal	We divided the parts of the proposals amongst the 3 teammates.	Complete proposal parts.
10/15/2024	Brealin, Yiyi, Xingjian	Group Chat (IM)	20 minutes	Review project proposals for turn in.	Reviewed the project proposal as a whole and prepared to turn it in.	Prepare for the next class and next project part.
10/23/2024	Brealin, Yiyi, Xingjian	In Person	80 minutes	Edit project proposals for resubmit	Redo the scope and functional requirement, add non-functional requirement section.	Review the proposal for resubmit.
11/4/2024	Brealin, Yiyi, Xingjian	In Person	40 Minutes	DFD and ERD	Developed our ERDs and reviewed our DFD	Continue reviewing to prepare to turn it in.
11/5/2024	Brealin, Yiyi, Xingjian	Group Chat (IM)	30 minutes	Review DFD and ERD, organize report materials	We organized the materials, finalized the report as a group.	It Would be good to turn it in.
11/18/2024	Brealin, Yiyi, Xingjian	In Person	40 minutes	Begin work on the final report.	Revising our DFD/ERD Being designing UI.	Continue to work on the final report.
12/1/2024	Brealin, Yiyi, Xingjian	IM Message	10 minutes	Presentation on Wednesday	Relay information from class to teammates.	Continue to work on the final report and presentation for Wednesday.
12/3/2024	Brealin, Yiyi, Xingjian	IM Message	30 minutes	Presentation on Wednesday	Checking in with each other before the presentations tomorrow. Worked on the presentation.	Prepare for the presentation tomorrow.